Treasury Inflation-Protected Securities (TIPS) and Breakeven Inflation - Reference Note

Reference note prepared as of 2026-06-30. This note is general market-context background on real-rate and inflation-expectations conventions; it is not part of, and does not feed, this engagement's credit or relative-value analysis of the target's own (nominal, fixed-rate) senior notes, which are priced entirely off the nominal Treasury and swap curves. It is included because a working analyst should be able to distinguish a real-rate instrument and its conventions from the nominal instruments this file's own target notes and comparables actually are, and because a breakeven curve is a common reference point in a broader market memo even when it is not itself a scored input here.

1. What a TIPS Is

A Treasury Inflation-Protected Security is a U.S. Treasury note or bond whose principal amount is adjusted for changes in the non-seasonally-adjusted Consumer Price Index for All Urban Consumers (CPI-U NSA). The coupon rate is fixed at issuance and is a real (inflation-adjusted) rate; the semiannual coupon payment is that fixed real rate applied to the current, inflation-adjusted principal, so the dollar coupon payment itself rises and falls with the inflation adjustment even though the coupon rate never changes. At maturity, the Treasury pays the greater of the inflation-adjusted principal or the original par amount - a deflation floor that protects a holder from a scenario of net cumulative deflation over the life of the security, a feature ordinary nominal Treasuries do not need because their principal is never adjusted in either direction.

The inflation adjustment itself uses a reference CPI value applying a short lag - the Treasury publishes a daily index ratio built from the CPI-U reading roughly three months prior, interpolated for daily granularity, so that the adjustment for any given day is known in advance rather than depending on a CPI release that has not happened yet. That lag is a structural feature of how TIPS are indexed, not a data quality issue, and it is the reason a TIPS holder's realized real return over any short window can differ slightly from the contemporaneous inflation rate.

2. Issuance and Auction Conventions

Treasury issues TIPS in 5-year, 10-year and 30-year original maturities on a regular auction calendar, with new issues and reopenings (additional issuance of an already-outstanding TIPS at the prevailing market price/yield rather than a newly set coupon) interspersed through the year, published in advance on Treasury's quarterly refunding calendar. Because the outstanding TIPS universe is materially smaller than the outstanding nominal Treasury universe, the on-the-run TIPS curve is less densely populated at every maturity point than the nominal par curve is, and a real-yield or breakeven reading at an off-the-run maturity commonly requires the same kind of curve interpolation this engagement's own nominal-curve analysis already uses for its own off-tenor points.

3. Real Yield vs. Nominal Yield

A TIPS' quoted yield is a real yield: the return an investor earns after inflation, because the principal itself is already being adjusted for realized inflation separately. A conventional Treasury note or bond's quoted yield is a nominal yield: the return before any inflation adjustment, because there is none. The two are not directly comparable without accounting for the inflation component embedded in the nominal yield, which is exactly what a breakeven reading is designed to isolate.

4. Breakeven Inflation: Definition and Approximation

The breakeven inflation rate at a given maturity is, informally, the average annual inflation rate that would make an investor indifferent between holding a nominal Treasury and holding a TIPS of the same maturity: if realized inflation over the life of the securities turns out to exceed the breakeven rate, the TIPS outperforms; if realized inflation falls short of the breakeven rate, the nominal bond outperforms. The commonly used approximation is the simple Fisher relation, nominal yield is approximately equal to real yield plus expected inflation plus a small inflation-risk/liquidity premium; rearranged, breakeven inflation is approximately nominal yield minus the TIPS real yield at the same maturity, with the caveat that the resulting figure blends genuine inflation expectations with a liquidity-and-risk-premium term that is not itself a market forecast and can move for reasons unrelated to inflation expectations (a TIPS liquidity squeeze during a market-stress episode, for instance, widens measured breakevens for reasons having nothing to do with what investors actually expect inflation to do).

A breakeven curve is therefore a genuinely useful, but genuinely approximate, market-implied inflation-expectations signal - useful for context in a broader market memo, and routinely quoted by the financial press and by Federal Reserve commentary alongside survey-based inflation expectations measures as one input among several, never as a precise point forecast on its own.

5. This Case: Scope and Non-Applicability

Nothing in this engagement's own valuation work uses a TIPS-derived real yield or a breakeven inflation rate as an input. The target's senior notes are a fixed-rate, nominal instrument, and this engagement's relative-value work (yield-to-worst, spread to the interpolated nominal Treasury curve, spread to the swap curve, option-adjusted spread) is entirely a nominal-curve exercise. This note exists for market-context completeness and so that a reader encountering TIPS or breakeven terminology elsewhere - in a broader macro commentary, for instance - has a correct, self-contained reference for what those terms mean and why they are not the same thing as the nominal-curve inputs this engagement's own analysis actually uses.

6. TIPS Program History

Treasury first issued TIPS in 1997, motivated by a combination of investor demand for an inflation hedge and the Treasury's own interest in a financing instrument that could lower its expected borrowing cost by not requiring

investors to price in a separate inflation-risk premium on top of expected inflation itself. The program's outstanding size grew steadily but has remained a minority share of total marketable Treasury debt outstanding across every year since introduction, reflecting a genuinely smaller natural investor base - primarily institutions and funds with an explicit real-return or inflation-hedging mandate - relative to the broad universe of investors that hold nominal Treasuries for duration, liquidity or reserve-asset purposes without any particular need for inflation protection specifically.

7. Why CPI-U NSA Specifically

TIPS index to the non-seasonally-adjusted CPI-U (Consumer Price Index for All Urban Consumers) rather than to core CPI, PCE, or a seasonally adjusted series. The choice of CPI-U NSA rather than a seasonally adjusted series is deliberate: a seasonally adjusted index is periodically revised as the Bureau of Labor Statistics updates its seasonal-adjustment factors, and an index used to set an investor's actual principal adjustment needs to be a stable, non-revised series so that a TIPS holder's principal adjustment on a given date is never later restated. The choice of headline CPI-U rather than a core measure (which excludes food and energy) reflects that TIPS are meant to hedge the inflation an investor actually experiences in the cost of living, not a policy-focused measure of underlying inflation trend that a central bank might prefer to target - the two purposes are related but distinct, and TIPS mechanics serve the former.

8. The Breakeven Curve's Shape and What Moves It

A breakeven curve, like a nominal yield curve, has a term structure: the 5-year breakeven, the 10-year breakeven and the 30-year breakeven need not move together, and their relative levels (a '5s10s30s' comparison, echoing the same curve-point shorthand used for the nominal curve) carry their own information. A steepening short-end breakeven relative to the long end, for instance, is commonly read as reflecting near-term inflation concerns without a corresponding change in longer-run inflation expectations, while a parallel shift across the whole breakeven curve is more consistent with a broad reassessment of inflation expectations at every horizon. Because a TIPS holder's realized return also depends on the liquidity-and-risk-premium term folded into the simple Fisher approximation (Section 4 above), a change in the breakeven curve's shape does not automatically mean the market has revised its inflation expectations - a widening TIPS liquidity premium during a period of market stress can move measured breakevens lower across the curve even if survey-based inflation expectations have not moved at all, which is exactly why breakeven readings are typically presented alongside, rather than in place of, survey-based inflation-expectations measures.

9. TIPS Liquidity Relative to Nominal Treasuries

The TIPS market, while deep by the standards of most fixed-income products, is meaningfully less liquid than the nominal Treasury market at a comparable maturity point - narrower dealer inventories, wider typical bid-ask spreads, and a smaller and less diverse holder base concentrated among real-money accounts with an explicit inflation mandate rather than the broad universe of official, bank and money-market holders that dominates nominal Treasury ownership. That liquidity gap is precisely the source of the liquidity-premium term the simple Fisher approximation in Section 4 above cannot separate from genuine inflation expectations, and it is the reason a sudden, broad flight-to-liquidity episode (a period of acute market stress, for instance) tends to widen TIPS-nominal spreads for reasons that have essentially nothing to do with the market's actual inflation outlook - investors sell the less liquid instrument first when they need cash quickly, regardless of what they think inflation will do.

10. Portfolio and Policy Uses, Briefly

Outside this engagement's own scope, TIPS and breakeven curves serve two broad, genuinely distinct purposes worth distinguishing. First, as a portfolio construction tool: an investor or plan sponsor with a real (inflation-linked) liability - a pension obligation with cost-of-living adjustments, for instance - can hold TIPS to hedge that liability's inflation sensitivity directly, in a way a nominal bond portfolio cannot. Second, as a policy and market-monitoring signal: the Federal Reserve and market commentary both routinely cite breakeven levels, and the shape of the breakeven curve, as one input (again, among several, and subject to the liquidity-premium caveat in Section 9 above) into an assessment of whether market-implied inflation expectations remain anchored near a central bank's target over the medium and long term.

11. Auctions, Bid-to-Cover, and Demand Signals

Each TIPS auction reports a bid-to-cover ratio - total bids received divided by the amount actually sold - together with the accepted (high) real yield and the shares awarded to different investor classes (primary dealers, direct bidders, indirect bidders, the last of which is commonly used as a rough proxy for foreign official and institutional demand, though not a precise one). A strong bid-to-cover ratio and a large indirect-bidder share are generally read as signs of healthy demand at the auctioned real yield; a weak auction, conversely, can itself move secondary-market real yields (and therefore breakevens) higher immediately afterward, independent of any new information about inflation expectations - another instance of the same general caution from Section 9 above, that not every observed move in a breakeven reading reflects a genuine change in the market's inflation view.

12. International Comparators

The United States is not the only sovereign issuer of inflation-linked debt. The United Kingdom's index-linked gilts (indexed to the UK's RPI historically, transitioning toward CPIH-alignment by the mid-2020s under a reform the UK authorities announced), France's OAT€i and French/euro-area OATei bonds (indexed to French CPI ex-tobacco and euro-area HICP ex-tobacco respectively), and Germany's inflation-linked Bunds are the most heavily referenced non-US comparators in cross-market inflation-linked analysis. Structural conventions - indexation lag length, deflation-

floor treatment, and which specific price index governs - differ by issuer, so a breakeven reading from one sovereign market is not a directly comparable figure to a breakeven reading from another without adjusting for those structural differences; this note limits itself to the US TIPS convention because that is the only inflation-linked structure potentially relevant to a USD-denominated engagement of this kind, and even then only as background, per Section 5 above.

13. Common Errors in Breakeven Analysis

Three errors recur often enough to flag explicitly. First, comparing a nominal yield and a TIPS real yield at mismatched maturities and calling the difference a breakeven rate - the comparison is only meaningful at a common maturity point, interpolated consistently on both curves. Second, treating the breakeven rate as a precise inflation forecast rather than as a market-implied figure carrying an embedded, time-varying liquidity-and-risk premium (Sections 4 and 9 above) - a breakeven move is genuine market information, but not the same thing as a clean read on expected inflation. Third, applying a nominal-curve day-count or compounding convention to a real-yield calculation, or vice versa, without adjusting - TIPS real-yield conventions and coupon mechanics differ from a conventional nominal note's in ways that matter for a precise calculation even though the general shape of the analysis is similar.

13a. TIPS Coupon Mechanics in More Detail

The semiannual coupon payment on a TIPS is the fixed real coupon rate, set at auction and unchanged for the life of the security, multiplied by the current inflation-adjusted principal on the relevant coupon date - not by the original par amount. Because the inflation-adjusted principal itself moves with the index ratio described in Section 1 above, the dollar coupon payment on a TIPS genuinely varies from period to period even though the coupon rate never resets, which is a source of periodic confusion for a reader used to a conventional fixed-rate note's constant dollar coupon. A TIPS' accrued interest calculation between coupon dates likewise applies the real coupon rate to the inflation-adjusted principal as of the settlement date, using the same daily index-ratio convention described in Section 1, so a TIPS trade's accrued-interest figure already embeds the inflation adjustment through settlement and is not a simple day-count fraction of a fixed dollar coupon the way a nominal note's accrued interest is.

13b. Tax Treatment, Briefly

For a US taxable holder, both the semiannual coupon payment and the annual increase in a TIPS' inflation-adjusted principal are generally treated as taxable income in the year accrued, even though the principal increase itself is not paid in cash until maturity or sale - the so-called 'phantom income' feature that leads many taxable investors to prefer holding TIPS in a tax-deferred account rather than a taxable one. This is a genuinely distinguishing feature relative to a nominal Treasury, whose only annual taxable cash-equivalent income is the coupon actually paid, and it is one more reason TIPS ownership patterns skew toward the institutional, real-money and tax-advantaged-account holder base described in Section 9 above rather than toward the broader universe of taxable retail holders.

13c. The TIPS Yield Curve vs. the Nominal Par Curve: A Worked Distinction

It is worth being explicit about a distinction that is easy to blur in casual usage: the nominal Treasury par yield curve this engagement's own relative-value work relies on (the real, non-monotonic 2026-06-30 curve used elsewhere in this file for the target notes' spread analysis) and a TIPS real-yield curve are two different curves built from two different sets of securities, and neither one is derived from the other. The nominal curve is built from conventional, non-inflation-indexed Treasury notes and bonds across the published tenor set; a TIPS real-yield curve is built entirely separately from the outstanding TIPS issues at their own respective maturities. A breakeven reading (Section 4 above) is the bridge between the two - it is computed FROM both curves, at a shared maturity point, rather than being a property of either curve standing alone. A reader who conflates 'the Treasury curve' generically, without specifying nominal or real, risks applying a real-yield number where a nominal-yield number belongs, or vice versa - exactly the error Section 3 above is written to forestall.

13d. Why This Note Exists Separately from the Nominal-Curve Analysis

This engagement's working documents reference the real, published, non-monotonic 2026-06-30 nominal Treasury par curve extensively, because that curve is a genuine input to the target notes' relative-value analysis. A reader moving quickly between documents in this file could reasonably wonder whether a TIPS or breakeven reference belongs in that same analytical chain. It does not, for the reasons stated plainly in Section 5 above, and this note is deliberately kept as a self-contained, separately filed reference specifically so that a reader who does not need it can skip it entirely, and a reader who does encounter TIPS or breakeven terminology elsewhere - in a piece of general market commentary attached to this file for context, for instance - has a single, correct place to look up what those terms actually mean without having to interrupt the nominal-curve analysis itself to do so.

14. Sources for This Note

The description above draws on publicly available TreasuryDirect and Bureau of the Fiscal Service program material on TIPS auction mechanics and indexation, Bureau of Labor Statistics documentation of the CPI-U series and its non-seasonally-adjusted variant, and Federal Reserve commentary on breakeven inflation as a market-based inflation-expectations measure. As with the companion SOFR transition note in this same folder, none of this note's content is derived from this engagement's own confidential transaction materials; it is general market-reference background only.

15. Glossary

Real yield: a bond's yield net of realized/expected inflation, i.e. a TIPS' quoted yield. Nominal yield: a bond's yield with

no inflation adjustment, i.e. a conventional Treasury's quoted yield. Breakeven inflation rate: nominal yield minus real yield at a common maturity, approximating market-implied average inflation expectations over that horizon, subject to an embedded risk/liquidity premium. Index ratio: the daily factor, built from a lagged CPI-U NSA reading, that scales a TIPS' original par principal up (or, in a deflation scenario, down before the maturity floor applies) to its current inflation-adjusted principal. Deflation floor: the guarantee that a TIPS holder receives at least original par principal at maturity, regardless of cumulative deflation. Bid-to-cover ratio: total auction bids received divided by the amount actually sold, a rough demand signal. Indirect bidder: an auction participant bidding through an intermediary rather than directly or as a primary dealer, commonly used as an imperfect proxy for institutional and foreign official demand.

16. Frequently Confused Terms

A short list of pairs that are easy to conflate but mean genuinely different things. Real yield vs. breakeven inflation: a real yield is a property of a single TIPS at a single maturity; a breakeven rate is a computed DIFFERENCE between a TIPS real yield and a nominal yield at a matched maturity, not a property of either bond alone (Section 4 above). CPI-U vs. CPI-U NSA: TIPS index specifically to the non-seasonally-adjusted series, not the more commonly quoted seasonally adjusted headline CPI-U figure most financial media report each month (Section 7 above) - the two track closely over a full year but can diverge materially month to month. Index ratio vs. inflation-adjusted principal: the index ratio is the scaling FACTOR; the inflation-adjusted principal is the RESULT of applying that factor to the original par amount (Section 1 above). Breakeven inflation vs. survey-based inflation expectations: the first is market-implied and priced continuously from two live curves; the second comes from periodic surveys of professional forecasters or consumers and is not itself a tradable-market figure - central bank commentary typically triangulates across both rather than relying on either alone (Section 10 above).